

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 1

Adaptive Classified

OL-T04 · Powers, Roots and Standard Form

With Answers

Questions with full worked solutions

■ 13 pages

FAMILY MAP

Powers Roots Standard Form

F01

Simplify numerical index expressions

6 Q | 10 marks

Question 24

4MA1-2H Q12 (a)

ORIGINAL QUESTION UNIT 1

2 marks

12 $2^7 \times 4^5 = 4^x$

(a) Calculate the value of x

$$x = \dots\dots\dots$$

(2)

Worked solution 24

Solve an index equation

MODULAR UNIT 1

2 marks

(a) – Solve an index equation

Write all terms in base 2

$$2^7 \cdot 4^5 = 2^7 \cdot (2^2)^5 = 2^{17}, \quad 4^x = (2^2)^x = 2^{2x}.$$

Equate exponents

$$2x = 17 \Rightarrow x = 8.5.$$

Final answer

$$x = 8.5$$

Question 08

4MA1-1H Q15

ORIGINAL QUESTION

3 marks

15 (a) $\sqrt{2} \div \frac{8^3}{16^2} = 2^n$

Work out the value of n
Show your working clearly.

$n = \dots\dots\dots$
(3)

Worked Solution 08

Numerical indices

MODULAR UNIT 1

3 marks

Write every term as a power of 2

$$\sqrt{2} = 2^{1/2}, \quad 8^3 = 2^9, \quad 16^{3/2} = 2^6.$$

Subtract indices

$$2^{1/2} \div 2^{9-6} = 2^{1/2-3} = 2^{-5/2}.$$

Final answer

$$n = -\frac{5}{2} = -2.5$$

1 (a) Simplify $e^8 \div e^2$

(1)

Worked solution

November 2021 | Question 1 | 1 marks

Family: Simplify numerical index expressions | Variant: Multiply or divide powers with a common base

METHOD / WORKING

$$1. (a) e^8 \div e^2 = e^6.$$

FINAL ANSWER

$$(a)e^6$$

- 1 (a) Write down the value of m , given that $3^4 \times 3^5 = 3^m$

$m = \dots\dots\dots$

(1)

Worked solution

November 2021 | Question 1 | 1 marks

Family: Simplify numerical index expressions | Variant: Multiply or divide powers with a common base

METHOD / WORKING

1. (a) $3^4 \times 3^5 = 3^9$, so $m = 9$.

FINAL ANSWER

 $m=9, n=21, p=4$

(b) Write down the value of n , given that $(5^3)^7 = 5^n$

$n = \dots\dots\dots$
(1)

Worked solution

November 2021 | Question 1 | 1 marks

Family: Simplify numerical index expressions | Variant: Power of a power or solve a numerical index relation

METHOD / WORKING

1. (b) $(5^3)^7 = 5^{21}$, so $n = 21$.

FINAL ANSWER

 $m=9, n=21, p=4$

(c) Find the value of p , given that $\frac{7^8 \times 7^2}{7^p} = 7^6$

$p = \dots\dots\dots$
(2)

Worked solution

November 2021 | Question 1 | 2 marks

Family: Simplify numerical index expressions | Variant: Power of a power or solve a numerical index relation

METHOD / WORKING

1. (c)

2. $\frac{7^8 \times 7^2}{7^p} = 7^{10-p} = 7^6$

3. So $10 - p = 6$, hence $p = 4$.4. Answer: $m = 9$, $n = 21$, $p = 4$ **FINAL ANSWER**

m=9, n=21, p=4